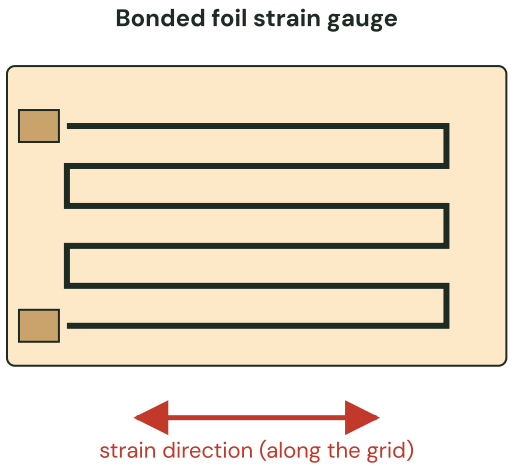
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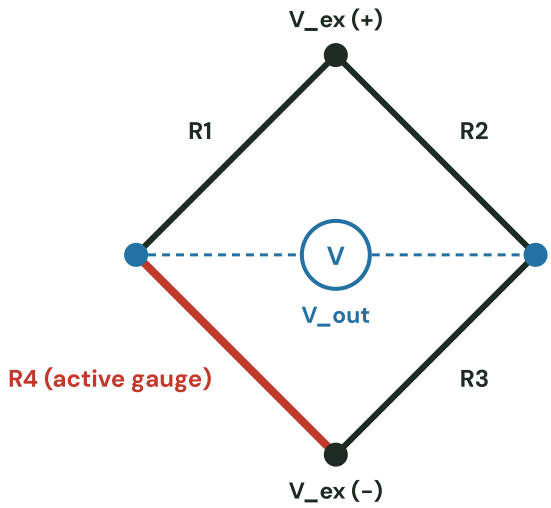
Strain Gauges — Theory and Use

Syllabus topic: Theory of strain gauges and their use

Principle

A bonded electrical-resistance strain gauge is a fine metal FOIL grid cemented to the surface. When the surface strains, the grid stretches or compresses and its RESISTANCE changes.





Bonded foil strain gauge and Wheatstone bridge (redrawn — the source figure labelled both supply nodes "V_in" and did not show where the output is taken)

$$\text{Gauge factor } GF = (\Delta R/R) \div (\Delta L/L) = (\Delta R/R) \div \varepsilon$$

Gauge	Gauge factor	Notes
METALLIC foil	≈ 2.0 (usually a little over 2)	Standard
SEMICONDUCTOR (piezoresistive)	≈ 20–40	Far more sensitive, but more temperature-dependent and non-linear

- Strain is tiny, so it is expressed in **MICROSTRAIN** (με = 10^{-6}).
- Common gauge resistances: **120 Ω** and **350 Ω**.

Example. GF = 2.0, ε = 500 με → ΔR/R = 2.0 × 500×10^{-6} = **0.001 (0.1%)** — only a **0.35 Ω** change on a 350 Ω gauge, far too small to read directly.

The Wheatstone bridge

Because ΔR is minute, gauges are connected in a WHEATSTONE BRIDGE that turns the small resistance change into a measurable voltage.

Bridge	Active gauges	Features
QUARTER	One	Basic; output V_out ≈ (V_ex/4)·GF·ε
HALF	Two	Doubled output; temperature compensation
FULL	Four	Maximum sensitivity; inherent temperature compensation

$$\text{Bridge balanced } (V_{out} = 0) \text{ when } R1/R2 = R4/R3$$

- A **DUMMY gauge** on an unstrained identical piece gives temperature compensation.
- A higher GF gives more output for the same strain.
- Strain gauges are the heart of LOAD CELLS, TORQUE sensors and PRESSURE transducers used for draft/force/power measurement (Units 2, 3, 6; Topic 5b).

Memorise: GF ≈ 2 (metal), 20–40 (semiconductor); GF = (ΔR/R)/ε; balance R1/R2 = R4/R3.

Practice MCQs

Q14. Gauge factor of a strain gauge is:

(a) (ΔL/L)/(ΔR/R) (b) (ΔR/R)/(ΔL/L)
(c) ΔR × ΔL (d) R/L

Q15. Typical gauge factor of a metallic foil strain gauge is about:

(a) 0.5 (b) 2.0
(c) 20 (d) 200

Q16. Semiconductor strain gauges have gauge factors around:

(a) 2 (b) 5
(c) 20–40 (d) 500

Q17. Strain is commonly expressed in:

(a) Newtons (b) Microstrain (10^{-6})
(c) Pascals (d) Ohms

Q18. For GF = 2 and ε = 500 με, ΔR/R equals:

(a) 0.0001 (b) 0.001
(c) 0.01 (d) 0.1

Q19. The small resistance change of a strain gauge is measured using a:

(a) Transformer (b) Wheatstone bridge
(c) Rectifier (d) Capacitor bank

Q20. A Wheatstone bridge is balanced when:

(a) R1 = R2 (b) R1/R2 = R4/R3
(c) All R zero (d) V_in = 0

Q21. Maximum bridge sensitivity and temperature compensation come from a:

(a) Quarter bridge (b) Half bridge
(c) Full bridge (four active gauges) (d) Single resistor

Q22. A dummy strain gauge is used for:

(a) Extra output (b) Temperature compensation
(c) Calibration weight (d) Grounding

Q23. Strain gauges are the sensing element of a:

(a) Thermometer (b) Load cell / torque sensor

(c) Tachometer (d) Flow meter only

Answers

Q14 — (b) (ΔR/R)/(ΔL/L) — *GF = (ΔR/R)/ε.*

Q15 — (b) 2.0 — *Metal ≈ 2.*

Q16 — (c) 20–40 — *Semiconductor 20–40.*

Q17 — (b) Microstrain (10^{-6})

Q18 — (b) 0.001 — *$\Delta R/R = GF\varepsilon = 2 \times 500\times10^{-6} = 0.001$.*

Q19 — (b) Wheatstone bridge

Q20 — (b) R1/R2 = R4/R3 — *Bridge balanced at R1/R2 = R4/R3.*

Q21 — (c) Full bridge (four active gauges) — *Full bridge = maximum sensitivity + temperature compensation.*

Q22 — (b) Temperature compensation — *A dummy gauge compensates temperature.*

Q23 — (b) Load cell / torque sensor

Sources: Doebelin — Measurement Systems; Beckwith — Mechanical Measurements; Sawhney — Electrical & Electronic Measurements; HBM/Omega strain-gauge technical notes (as given in the source notes, July 2026); ASAE instrumentation standards; NPTEL measurement lectures.